# Introduction to Linux Systems

## Bash and Shell Scripts

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# Lab1: Personalize Primary Prompt

- Change the original style of the primary prompt with the following requirements:
  - New color
  - Extend with your student id
  - Replace “$” with “#”

- Example:

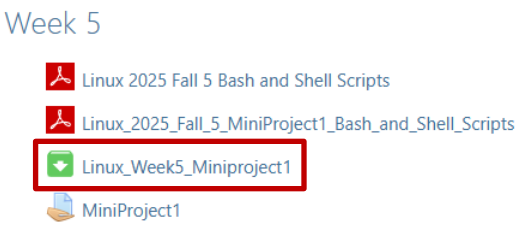
```
liunx@VirtualBox:~$
```

⇨

```
liunx@VirtualBox-P12345678:~#
```

# Lab2: Automated Uncompressing

- The given directory “Linux_Week5_Miniproject1” has the following contents:




**Linux_Week5_Miniproject1/**

- **compressed_files/ (all .zip .rar .tar.gz and unknown files)**
- **student_id (the list of student id for uncompressing files)**

```
tyt@tyt-vm:~/Downloads/Linux_Week5_Miniproject1$ ls
compressed_files  student_id
```

# Lab2: Automated Uncompressing

- Giving a bunch of compressing files named with student id, the files will be **.zip**, **.tar.gz**, **.rar** and **unknown**.
- Using the student ID list to find out
  - who did not hand in files. (record into **missing_list**)
  - who hand in the incorrect file format. (record into **wrong_list**)

```
tyt@tyt-vm:~/Downloads/Linux_Week5_Miniproject1/compressed_files$ ls
P11013888.zip      P32778596.zip      P50805555.dika     P72319701.rar
P11658996.zip      P33382480.zip      P51718038.tar.gz   P72857984.zip
P13080713.rar      P34949798.tar.gz   P52533795.dika     P74266531.dika
P17055460.zip      P36206144.zip      P54554905.rar      P74696826.tar.gz
P17475677.rar      P40957552.zip      P55006021.rar      P76651060.zip
P17781209.rar      P41912439.dika     P56259680.zip      P77016520.zip
P19313522.tar.gz   P42311654.tar.gz   P58085617.rar      P77383522.tar.gz
P19344950.tar.gz   P42435480.zip      P58125242.tar.gz   P77756449.rar
P22039141.rar      P43334372.zip      P58762070.tar.gz   P78983131.dika
P22352956.zip      P43417222.tar.gz   P59054121.rar      P79442744.zip
P22820041.rar      P45144248.zip      P59361378.tar.gz   P83169523.dika
P23311839.dika     P46090767.dika     P61310536.zip      P83573447.dika
P23951120.zip      P46461544.zip      P61381026.tar.gz   P87697607.dika
P24178422.tar.gz   P47636709.rar      P61745478.tar.gz   P90116193.rar
P24906604.zip      P47919394.tar.gz   P62830429.rar      P90837989.rar
P24978236.zip      P48229362.tar.gz   P63312965.rar      P91729589.rar
P26705550.tar.gz   P48703567.dika     P64686240.zip      P92387984.zip
P26862150.tar.gz   P48727828.zip      P65482877.rar      P97773727.dika
```

```
tyt@tyt-vm:~/Downloads/Linux_Week5_Miniproject1$ cat student_id
P65482877
P47919394
P67547151
P78983131
P50805555
P61745478
P46090767
P19344950
P62830429
P30965598
P52533795
P74266531
P27451331
P63312965
P97773727
P87697607
P36206144
```

# Lab2: Automated Uncompressing

- The files will be .zip, .tar.gz, .rar and unknown, please classify them into four groups.
  - Uncompressing tool: $ **sudo apt install tar unzip unrar**
- Finally, uncompress them all except unknown files.

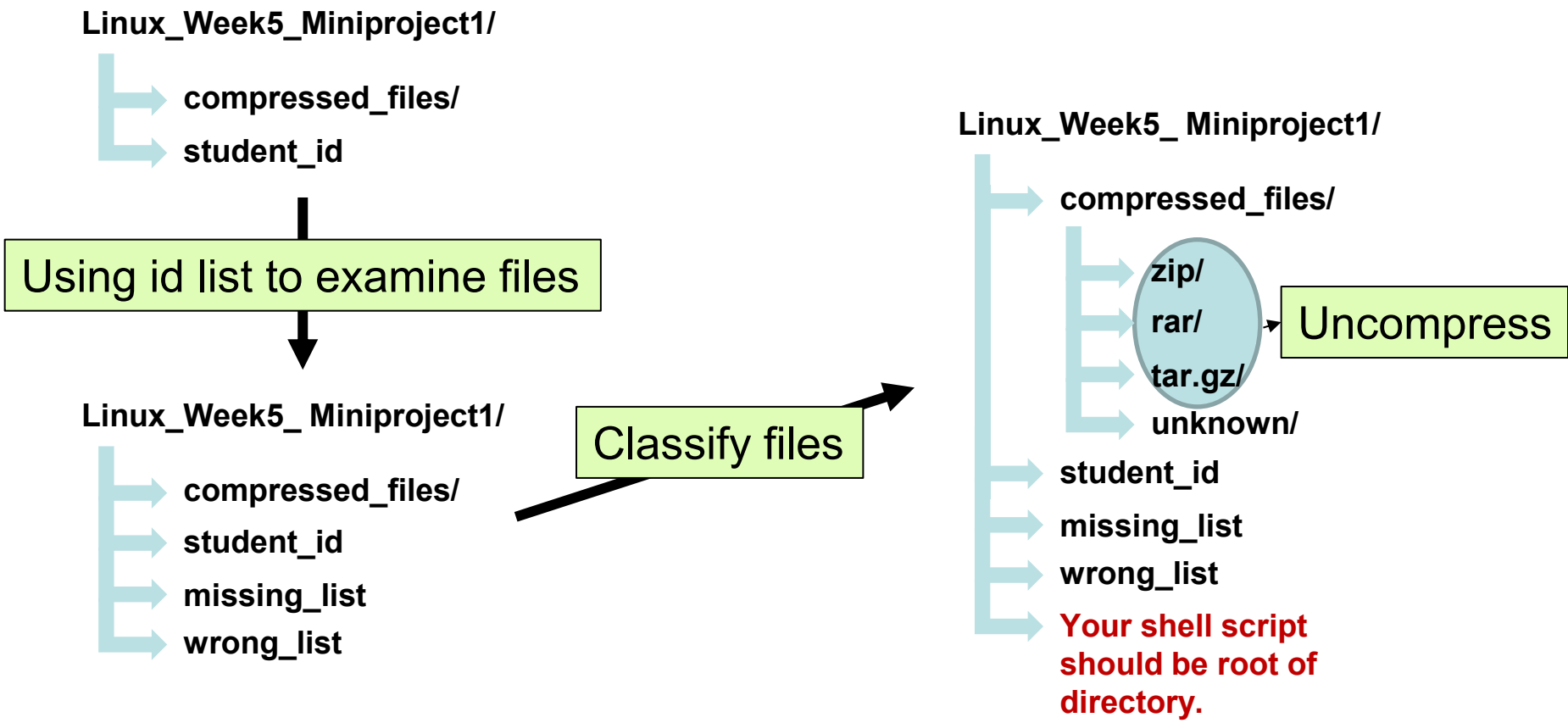
```
tyt@tyt-vm:~/Downloads/Linux_Week5_Miniproject1/compressed_files$ ls
P11013888.zip      P32778596.zip      P50805555.dika     P72319701.rar
P11658996.zip      P33382480.zip      P51718038.tar.gz   P72857984.zip
P13080713.rar      P34949798.tar.gz   P52533795.dika     P74266531.dika
P17055460.zip      P36206144.zip      P54554905.rar      P74696826.tar.gz
P17475677.rar      P40957552.zip      P55006021.rar      P76651060.zip
P17781209.rar      P41912439.dika     P56259680.zip      P77016520.zip
P19313522.tar.gz   P42311654.tar.gz   P58085617.rar      P77383522.tar.gz
P19344950.tar.gz   P42435480.zip      P58125242.tar.gz   P77756449.rar
P22039141.rar      P43334372.zip      P58762070.tar.gz   P78983131.dika
P22352956.zip      P43417222.tar.gz   P59054121.rar      P79442744.zip
P22820041.rar      P45144248.zip      P59361378.tar.gz   P83169523.dika
P23311839.dika     P46090767.dika     P61310536.zip      P83573447.dika
P23951120.zip      P46461544.zip      P61381026.tar.gz   P87697607.dika
P24178422.tar.gz   P47636709.rar      P61745478.tar.gz   P90116193.rar
P24906604.zip      P47919394.tar.gz   P62830429.rar      P90837989.rar
```

After sorting

```
tyt@tyt-vm:~/Downloads/Linux_Week5_Miniproject1/compressed_files$ ls
rar  tar.gz  unknown  zip
```

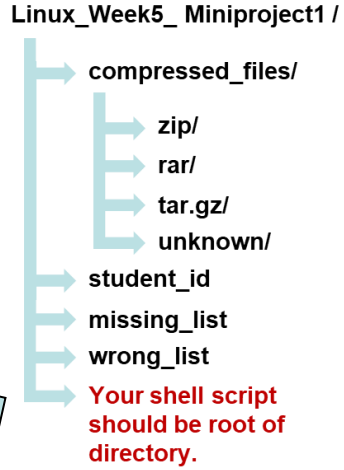
# Lab2: Automated Uncompressing

- The flow of overall steps and final directory tree.



# Video Demo

- Due Date (Upload in Moodle): **12:00 on 10/29 Wed.**
- The .zip file should contain **one video, Linux_Week5_Miniproject1 folder and the text file of thoughts & suggestions**.
- Lab1 (15%)
  - Show how you edit the Primary Prompt in the video.
- Lab2 (80%)
  - Show how your shell script works and explain the detail of your code in the video. (do not exceed 5 minutes)
  - Update Linux_Week5_Miniproject1.
- Thoughts & suggestions (5%)
- **Wrong folder structure will lose 10 points.**




格式: Miniproject1_*StudentID*.zip
e.g. Miniproject1_F12345678.zip

# QUESTIONS